

↑ Prove that if  $\{a_n\}$  is sequence of integers s.t. infinitely many non-zero terms then the convergence bound of  $\sum a_n$  is at most 1.

Sol). If  $a_n = 1 \forall n \in \mathbb{N}$ , then  $R = 1$ .

And, by the theorem,  $R > 1 \Leftrightarrow \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$ .

Put  $\limsup \sqrt[n]{|a_n|} = \beta < 1$ . Then, for  $\beta < \alpha < 1$ ,

$$\exists N \in \mathbb{N} \text{ s.t. } n \geq N \Rightarrow \sqrt[n]{|a_n|} < \alpha \\ \Leftrightarrow |a_n| < \alpha^n < 1$$

Contradiction.

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Let  $a_n > 0$ ,  $S_n = a_1 + \dots + a_n$ , and  $\sum a_n$  diverges. Prove that

a)  $\sum \frac{a_n}{1+a_n}$  diverges.

b)  $\sum \frac{a_n}{S_n}$  diverges.

$$a_n \leq \frac{1}{2}(1+a_n)$$

$$\Rightarrow a_n \leq 1$$

c)  $\sum \frac{a_n}{S_n^2}$  converges.

d)  $\sum \frac{a_n}{1+a_n}$  and  $\sum \frac{a_n}{1+n^2 a_n}$  converges?

Sol a) Suppose that  $\{a_n\}$  is bounded by  $M > 0$ . Then,  $a_n < M$  implies  $\frac{1}{1+a_n} > \frac{1}{1+M}$ .

$$\text{Therefore } \sum_{n=1}^m \frac{a_n}{1+a_n} > \sum_{n=1}^m \frac{a_n}{1+M} = \frac{1}{1+M} \cdot \sum_{n=1}^m a_n$$

diverges.

And, if  $\{a_n\}$  is not bounded, then there are infinitely many  $n \in \mathbb{N}$  s.t.  $a_n > 1$ .

This implies that  $\frac{a_n}{1+a_n} > \frac{a_n}{a_n+a_n} = \frac{1}{2}$  for infinitely many  $n$ . Therefore diverges.

Alternatively, using contraposition: Suppose  $\sum \frac{a_n}{1+a_n}$  converges. Take  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow \frac{a_n}{1+a_n} < \frac{1}{2}$$

It equals to  $a_n < 1$ . Now, for  $n \geq N$ ,  $\frac{a_n}{2} \leq \frac{a_n}{1+a_n}$  implies  $\sum \frac{a_n}{2}$  converges.

(Thanks to 3.1)

Sol b). For any  $N \in \mathbb{N}$  and  $K \geq 0$ ,

$$\frac{a_{N+1}}{S_{N+1}} + \dots + \frac{a_{N+K}}{S_{N+K}} \geq \frac{1}{S_{N+K}} \cdot (a_{N+1} + \dots + a_{N+K}) = \frac{S_{N+K} - S_N}{S_{N+K}} = \left(1 - \frac{S_N}{S_{N+K}}\right) \quad (*)$$

If  $\sum \frac{a_n}{S_n}$  converges, then there exists  $N \in \mathbb{N}$  s.t.

$$m \geq n \geq N \Rightarrow \left| \sum_{k=n}^m \frac{a_k}{S_k} \right| < \frac{1}{2}.$$

But, since the inequality (\*), and  $S_n = \sum_{i=1}^n a_i$  diverges, we can choose  $K \geq N$  s.t.

$$\frac{1}{2} \leq 1 - \frac{S_N}{S_{N+K}} \leq \sum_{i=N+1}^{N+K} \frac{a_i}{S_i}$$

Which is contradiction.

Sol c) Since  $s_{n-1} < s_n$  for any  $n \in \mathbb{N}$ ,

$$\frac{a_n}{s_n^2} \leq \frac{a_n}{s_{n-1} s_n} = \frac{s_n - s_{n-1}}{s_{n-1} s_n} = \frac{1}{s_{n-1}} - \frac{1}{s_n} \quad (n \geq 2)$$

Now, for any  $n \geq 2$ ,

$$\sum_{k=2}^n \frac{a_k}{s_k^2} \leq \sum_{k=2}^n \frac{1}{s_{k-1}} - \frac{1}{s_k} = \frac{1}{s_1} - \frac{1}{s_n} \rightarrow \frac{1}{s_1} \quad \text{as } n \rightarrow \infty$$

Since  $s_n$  diverges to  $\infty$ ,

Sol d) First,

$$\frac{a_n}{1+n^2 a_n} \leq \frac{a_n}{n^2 a_n} = \frac{1}{n^2}, \quad \text{therefore } \sum \frac{a_n}{1+n^2 a_n} \leq \sum \frac{1}{n^2}, \text{ converges.}$$

To show the second assertion,

if  $\{na_n\}$  is bounded by  $M > 0$ , then

$$na_n < M \text{ implies } \frac{a_n}{1+na_n} > \frac{a_n}{1+M}, \quad \text{therefore } \sum \frac{a_n}{1+na_n} > \frac{1}{1+M} \cdot \sum a_n \text{ diverges.}$$

if  $\{na_n\}$  is not bounded, then we can not determine convergence. For instance, define

$$a_n = \begin{cases} 1 & \text{if } n=k^2 \text{ for some } k \in \mathbb{N} \\ \frac{1}{n^2} & \text{otherwise} \end{cases}$$

Then,  $\sum a_n$  diverges since  $a_n$  does not converge to 0,

but  $\sum \frac{a_n}{1+na_n}$  converges, because

$$\frac{a_n}{1+na_n} = \begin{cases} \frac{1}{1+n} & \text{if } n=k^2 \text{ for some } k \in \mathbb{N} \quad \left( < \frac{1}{n} \right) \\ \frac{1}{n^2+n} & \text{otherwise} \quad \left( < \frac{1}{n^2} \right) \end{cases}$$

which satisfies  $\sum_{n=1}^m \frac{a_n}{1+na_n} \leq \sum_{n=1}^m \frac{1}{n^2} < \infty$  for all  $m \in \mathbb{N}$ .

Meanwhile, put  $a_n = \frac{1}{n}$ . Then both

$$\sum a_n, \quad \sum \frac{a_n}{1+na_n} = \sum \frac{1}{2n}$$

are divergent.

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Let  $a_n > 0$ ,  $\sum a_n$  converges, put  $r_n = \sum_{m=n}^{\infty} a_m$

a). Prove  $\forall N, k \in \mathbb{N}$ ,

$$\frac{a_N}{r_N} + \dots + \frac{a_{N+k}}{r_{N+k}} > 1 - \frac{r_{N+k}}{r_N}$$

Deduce that  $\sum \frac{a_n}{r_n}$  diverges.

$$\frac{r_N - a_N}{r_N} = \frac{r_{N+1}}{r_N}$$

b). Prove  $\forall n \in \mathbb{N}$ ,

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Deduce that  $\sum \frac{a_n}{\sqrt{r_n}}$  converges.

Sol). Since  $\sum a_n$  converges, put  $A = \sum_{n=1}^{\infty} a_n$ . Then,

$$A - r_n = \sum_{k=1}^{n-1} a_k$$

$r_n \rightarrow 0$  ?

And,  $r_n$  is decreasing. Now,  $r_N > r_{N+k} \Rightarrow \frac{1}{r_N} < \frac{1}{r_{N+k}}$  implies

$$\begin{aligned} \frac{a_N}{r_N} + \dots + \frac{a_{N+k}}{r_{N+k}} &> \frac{1}{r_N} (a_N + a_{N+1} + \dots + a_{N+k}) \\ &= \frac{1}{r_N} (r_N - r_{N+k+1}) \\ &= 1 - \frac{r_{N+k+1}}{r_N} > 1 - \frac{r_{N+k}}{r_N} \end{aligned}$$

Now, suppose  $\sum \frac{a_n}{r_n}$  converges. Then, there exists  $N \in \mathbb{N}$  s.t. for all  $k \in \mathbb{N}$ ,

$$\left| \sum_{n=N}^{N+k} \frac{a_n}{r_n} \right| < \frac{1}{2}.$$

Meanwhile,  $\lim_{n \rightarrow \infty} r_n = 0$  because for any  $\epsilon > 0$ , we can choose  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow |r_n| = r_n = \sum_{m=n}^{\infty} a_m = A - \sum_{k=1}^{n-1} a_k < \epsilon$$

Therefore, we can choose  $K \in \mathbb{N}$  for fixed  $N$ ,

$$k \geq K \Rightarrow \left| \sum_{n=N}^{N+k} \frac{a_n}{r_n} \right| > 1 - \frac{r_{N+k}}{r_N} \geq \frac{1}{2}.$$

It is contradiction.

Sol b). Since  $a_n = r_n - r_{n+1} = (\sqrt{r_n} - \sqrt{r_{n+1}})(\sqrt{r_n} + \sqrt{r_{n+1}})$  ( $n \geq 1$ )

$$\frac{a_n}{\sqrt{r_n}} = (\sqrt{r_n} - \sqrt{r_{n+1}}) \left(1 + \frac{\sqrt{r_{n+1}}}{\sqrt{r_n}}\right)$$

$$< (\sqrt{r_n} - \sqrt{r_{n+1}})(1 + 1) = 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Therefore,

$$\sum_{k=1}^n \frac{a_k}{\sqrt{r_k}} < \sum_{k=1}^n 2(\sqrt{r_k} - \sqrt{r_{k+1}})$$

$$= 2(\sqrt{r_1} - \sqrt{r_{n+1}}) \rightarrow 2\sqrt{r_1} \text{ as } n \rightarrow \infty.$$

Thus  $\sum \frac{a_n}{\sqrt{r_n}}$  converges.

Section 3.13

Given abs. convergent  $\sum a_n, \sum b_n$ ,  $\sum_{r=0}^n \sum_{k=1}^n a_{n-k} b_k$  converges absolutely.

Proof. Denote  $A_n = \sum_{k=0}^n |a_k|$ ,  $B_n = \sum_{k=0}^n |b_k|$ , and  $C_n = \sum_{k=0}^n \left| \sum_{r=0}^k a_{k-r} b_r \right|$ ,  $\beta_n = B - B_n$

Put  $A = \lim_{n \rightarrow \infty} A_n$ ,  $B = \lim_{n \rightarrow \infty} B_n$

$$C_n = |a_0 b_0| + |a_1 b_0 + a_0 b_1| + |a_2 b_0 + a_1 b_1 + a_0 b_2| + \dots + |a_n b_0 + \dots + a_0 b_n|$$

$$\leq |a_0| B_n + |a_1| B_{n-1} + \dots + |a_n| B_0$$

$$= |a_0| (B - \beta_n) + |a_1| (B - \beta_{n-1}) + \dots + |a_n| (B - \beta_0)$$

$$= A_n B - (|a_0| \beta_n + |a_1| \beta_{n-1} + \dots + |a_n| \beta_0)$$

$$\leq A_n B \rightarrow AB$$

Therefore,  $C_n$  is monotone increasing bounded sequence, it is converges.

Section 3.14.

Let  $\{s_n\}$  be a complex numbers sequence, and define

$$\sigma_m = \frac{s_1 + s_2 + \dots + s_m}{m} \quad (m \in \mathbb{N})$$

a. If  $\lim_{n \rightarrow \infty} s_n = s$ , then  $\lim_{n \rightarrow \infty} \sigma_n = s$

sol) Given  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow |s_n - s| < \frac{\varepsilon}{2}$$

meanwhile, observe that for  $m > N$ ,

$$\begin{aligned} |m\sigma_m - ms| &= |(s_1 - s) + \dots + (s_N - s) + (s_{N+1} - s) + \dots + (s_m - s)| \\ &\leq |(s_1 - s) + \dots + (s_N - s)| + |s_{N+1} - s| + \dots + |s_m - s| \\ &\leq |(s_1 - s) + \dots + (s_N - s)| + m \cdot \frac{\varepsilon}{2} \quad (m \cdot \frac{\varepsilon}{2} \text{ can be replaced by } (m-N) \frac{\varepsilon}{2}) \end{aligned}$$

Taking  $\frac{1}{m}$  both side, then

$$|\sigma_m - s| \leq \frac{1}{m} \cdot |(s_1 - s) + \dots + (s_N - s)| + \frac{\varepsilon}{2}$$

now, Take  $M \in \mathbb{N}$  s.t.  $|s_1 - s| + \dots + |s_N - s| < M \cdot \frac{\varepsilon}{2}$ , then

$$m \geq M \Rightarrow |\sigma_m - s| < \varepsilon.$$

a-1) If  $\{s_n\}$  is a real numbers, then

$$\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} \sigma_n \leq \limsup_{n \rightarrow \infty} \sigma_n \leq \limsup_{n \rightarrow \infty} s_n$$

proof. Given  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow s_n < \limsup_{n \rightarrow \infty} s_n + \frac{\varepsilon}{2}$$

$$\begin{aligned} \sigma_n &= \frac{s_1 + \dots + s_N + s_{N+1} + \dots + s_n}{n} = \frac{1}{n} (s_1 + \dots + s_N) + \frac{1}{n} (s_{N+1} + \dots + s_n) \\ &\leq \frac{1}{n} (s_1 + \dots + s_N) + \frac{1}{n} \cdot (n-N) \cdot \left( \limsup_{n \rightarrow \infty} s_n + \frac{\varepsilon}{2} \right) \\ &\leq \frac{1}{n} (s_1 + \dots + s_N) + \limsup_{n \rightarrow \infty} s_n + \frac{\varepsilon}{2} \end{aligned}$$

Take  $M \in \mathbb{N}$  s.t.  $(s_1 + \dots + s_N) < M \cdot \frac{\varepsilon}{2}$ . Then, for  $n \geq M$ ,

$$\sigma_n \leq \limsup_{n \rightarrow \infty} s_n + \varepsilon.$$

Since  $\varepsilon > 0$  was chosen arbitrarily,  $\limsup_{n \rightarrow \infty} \sigma_n \leq \limsup_{n \rightarrow \infty} s_n$ .

d). put  $a_n = s_{n+1} - s_n$ ,  $n \geq 1$ . Show that

$$s_{n+1} - s_n = \frac{1}{n} \cdot \sum_{k=1}^n k a_k$$

And deduce that  $\lim_{n \rightarrow \infty} n a_n = 0$  and  $\lim_{n \rightarrow \infty} s_n = s$  implies  $\lim_{n \rightarrow \infty} s_n = s$ .

Proof. Since

$$\begin{aligned} \sum_{k=1}^n k a_k &= a_1 + 2a_2 + 3a_3 + \dots + n a_n \\ &= (s_2 - s_1) + 2(s_3 - s_2) + \dots + n(s_{n+1} - s_n) \\ &= n s_{n+1} - s_n - s_{n-1} - \dots - s_1 \\ &= n s_{n+1} - (s_1 + \dots + s_n) \\ &= n s_{n+1} - n \cdot \sigma_n \end{aligned}$$

Therefore,

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left( \sigma_n + \underbrace{\frac{1}{n} \sum_{k=1}^n k a_k}_{=0, \text{ by a)}} = \lim_{n \rightarrow \infty} \sigma_n = s.$$

e) Assume  $\{n a_n\}$  bounded and  $\lim_{n \rightarrow \infty} s_n = s$ , then  $\lim_{n \rightarrow \infty} s_n = s$ .

Proof. Let  $M > 0$  be a bound of  $n a_n$ .

Observe that for  $n > m$ ,

$$(n > m, \text{ then } \frac{1}{n} = \frac{m}{n-m} \left( \frac{1}{m} - \frac{1}{n} \right),$$

$$\text{it is derived by } \frac{1}{AB} = \frac{1}{B-A} \left( \frac{1}{A} - \frac{1}{B} \right)$$

$$\begin{aligned} s_n - \sigma_n &= s_n - \frac{1}{n} (s_1 + \dots + s_n) \\ &= s_n - \frac{m}{n-m} \left( \frac{1}{m} - \frac{1}{n} \right) (s_1 + \dots + s_n) \\ &= s_n - \frac{m}{n-m} \left( \frac{1}{m} (s_1 + \dots + s_m + s_{m+1} + \dots + s_n) - \frac{1}{n} (s_1 + \dots + s_n) \right) \\ &= s_n - \frac{m}{n-m} \left( \frac{1}{m} [m \sigma_m + (s_{m+1} + \dots + s_n)] - \sigma_n \right) \\ &= s_n - \frac{m}{n-m} \left( \sigma_m - \sigma_n + \frac{1}{m} [s_{m+1} + \dots + s_n] \right) \\ &= \frac{m}{n-m} (\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{k=m+1}^n (s_n - s_k) \end{aligned}$$

$$\begin{aligned} -1 + m\varepsilon \leq n &\Leftrightarrow \frac{m}{n-m+1} \leq \frac{1}{\varepsilon} \\ n - \varepsilon < m + m\varepsilon &\Leftrightarrow \frac{n-m}{m} < \varepsilon \end{aligned}$$

Meanwhile, since  $|n a_n| = |n(s_{n+1} - s_n)| < M$ , for  $i \geq m+1$ ,

$$\begin{aligned} |s_n - s_i| &= |s_n - s_{n-1} + s_{n-1} - s_{n-2} + s_{n-2} - \dots + s_{i+1} - s_i| \\ &\leq |s_n - s_{n-1}| + |s_{n-1} - s_{n-2}| + \dots + |s_{i+1} - s_i| \\ &\leq \frac{M}{n-1} + \frac{M}{n-2} + \dots + \frac{M}{i} \leq \frac{n-i}{i} \cdot M \leq \frac{n-m}{m+1} \cdot M \leq \frac{n-m}{m+1} \cdot M. \end{aligned}$$

Given  $\varepsilon > 0$  and for each  $n \in \mathbb{N}$ , take  $m \in \mathbb{N}$  satisfying

$$m \stackrel{(1)}{\leq} \frac{n-\varepsilon}{1+\varepsilon} \stackrel{(2)}{<} m+1$$

Then,  $m < n$  and  $(1) \Leftrightarrow m + m\varepsilon + \varepsilon \leq n \Leftrightarrow \frac{m+1}{n-m} \leq \frac{1}{\varepsilon}$

and  $(2) \Leftrightarrow n - \varepsilon < m + 1 + \varepsilon \stackrel{(1)}{\Leftrightarrow} \frac{n-m}{m} < \varepsilon$  implies if  $i \geq m+1$ ,  $|s_n - s_i| \leq \frac{n-m}{m+1} \cdot M < \varepsilon M$ .

$$\begin{aligned} \text{Finally } \limsup_{n \rightarrow \infty} |s_n - s| &\leq \limsup_{n \rightarrow \infty} |s_n - \sigma_n| + \limsup_{n \rightarrow \infty} |\sigma_n - s| = \limsup_{n \rightarrow \infty} |s_n - \sigma_n| \\ &\leq \limsup_{n \rightarrow \infty} \left( \frac{m}{n-m} |\sigma_n - \sigma_m| \right) + \limsup_{n \rightarrow \infty} \left( \frac{1}{n-m} \sum_{k=m+1}^n |s_n - s_k| \right) \\ &\leq \frac{1}{\varepsilon} \limsup_{n \rightarrow \infty} |\sigma_n - \sigma_m| + \varepsilon M = \varepsilon M \end{aligned}$$

Since  $\varepsilon > 0$  was arbitrary,  $\lim_{n \rightarrow \infty} s_n = s$ .

Section 3.16.

Fix  $\alpha > 0$ , and choose  $x_1 > \sqrt{\alpha}$ . Define  $\{x_n\}$  as recursively:

$$x_{n+1} := \frac{1}{2} \cdot \left( x_n + \frac{\alpha}{x_n} \right)$$

a).  $\{x_n\}$  decreases monotonically, and  $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$

Proof. observe that

$$x_{n+1} \leq x_n \Leftrightarrow \alpha \leq x_n^2 \Leftrightarrow \frac{\alpha}{x_n} \leq x_n$$

For  $n=1$ , it is trivially true. Suppose for  $n \in \mathbb{N}$ , it is true. Then,

$$x_{n+1} = \frac{1}{2} \cdot \left( x_n + \frac{\alpha}{x_n} \right) \leq \frac{1}{2} \cdot \left( x_n + x_n \right) = x_n. \quad \text{Point: } \frac{\alpha}{x_n} \leq x_n$$

By induction,  $\{x_n\}$  is monotone decreasing. And  $x_n \geq \sqrt{\alpha}$  for all  $n \in \mathbb{N}$ .

$\{x_n\}$  is decreasing bounded, therefore converges. Put  $x = \lim_{n \rightarrow \infty} x_n \geq \sqrt{\alpha}$ . Now,

$$x = \lim_{n \rightarrow \infty} x_{n+1} = \frac{1}{2} \left( \lim_{n \rightarrow \infty} x_n + \frac{\alpha}{\lim_{n \rightarrow \infty} x_n} \right) = \frac{1}{2} \left( x + \frac{\alpha}{x} \right) \Rightarrow x = \sqrt{\alpha}.$$

b) Put  $e_n = x_n - \sqrt{\alpha}$  (That is, error between  $x_n$  and  $\sqrt{\alpha}$ , magnitude for approximation)

Show that  $e_{n+1} = \frac{e_n^2}{2x_n} < \frac{e_n^2}{2\sqrt{\alpha}}$  and that  $e_{n+1} < 2\sqrt{\alpha} \cdot \left( \frac{e_1}{2\sqrt{\alpha}} \right)^{2^n}$  for  $n \in \mathbb{N}$

Proof since

$$e_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left( x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \frac{e_n^2 + 2e_n\sqrt{\alpha} + 2\alpha}{x_n} - \sqrt{\alpha} = \frac{e_n^2}{2x_n} + \frac{\sqrt{\alpha}}{x_n} \cdot (e_n + \sqrt{\alpha}) - \sqrt{\alpha} = \frac{e_n^2}{2x_n}$$

and  $x_n \geq \sqrt{\alpha}$  gives  $e_{n+1} < \frac{e_n^2}{2\sqrt{\alpha}}$ . Now, inductively,

$$\begin{aligned} e_{n+1} &< \frac{e_n^2}{2\sqrt{\alpha}} < \frac{1}{2\sqrt{\alpha}} \cdot \left( \frac{e_{n-1}^4}{4\alpha} \right) = \frac{1}{2\sqrt{\alpha}} \cdot \frac{1}{(2\sqrt{\alpha})^2} \cdot e_{n-1}^4 < \dots \left( \sum_{k=0}^{n-1} 2^k = 2^n - 1 \right) \\ &< \left( \frac{1}{2\sqrt{\alpha}} \right)^{1+2+2^2+\dots+2^{n-1}} \cdot e_{n-(n-1)}^{2^n} = 2\sqrt{\alpha} \cdot \left( \frac{e_1}{2\sqrt{\alpha}} \right)^{2^n} \end{aligned}$$



17. Fix  $\alpha > 1$ , take  $x_1 > \sqrt{\alpha}$ . Define recursively

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}$$

a, b) Prove  $x_1 > x_3 > x_5 > \dots$  and  $x_2 < x_4 < x_6 < \dots$

Proof. Observe that for  $n \in \mathbb{N}$ ,  $x_{n+1} = \frac{\alpha + x_n}{1 + x_n} > \frac{1 + x_n}{1 + x_n} = 1$ . And

$$x_{n+1}^2 = \frac{\alpha^2 + 2\alpha x_n + x_n^2}{1 + 2x_n + x_n^2} = \frac{\alpha(1 + 2x_n + x_n^2) + \alpha^2 - \alpha - (\alpha - 1)x_n^2}{1 + 2x_n + x_n^2} = \alpha + (\alpha - 1) \cdot \frac{\alpha - x_n^2}{1 + 2x_n + x_n^2}$$

Therefore,  $\begin{cases} x_{n+1} < \sqrt{\alpha} & \text{if } x_n > \sqrt{\alpha} \\ x_{n+1} > \sqrt{\alpha} & \text{if } x_n < \sqrt{\alpha} \end{cases}$  implies  $\begin{cases} x_{2n} < \sqrt{\alpha} \\ x_{2n-1} > \sqrt{\alpha} \end{cases}$  for all  $n \in \mathbb{N}$ .

$$\text{Meanwhile, } x_{n+2} = \frac{\alpha + x_{n+1}}{1 + x_{n+1}} = \frac{\alpha + \frac{\alpha + x_n}{1 + x_n}}{1 + \frac{\alpha + x_n}{1 + x_n}} = \frac{2\alpha + (\alpha + 1)x_n}{2x_n + \alpha + 1} = x_n \cdot \frac{2 \cdot \frac{\alpha}{x_n} + \alpha + 1}{2x_n + \alpha + 1}$$

For  $n = 2k$ ,  $x_{2k} < \sqrt{\alpha}$  implies  $x_{2k} < \frac{\alpha}{x_{2k}}$  and

$n = 2k-1$ ,  $x_{2k-1} > \sqrt{\alpha}$  implies  $x_{2k-1} > \frac{\alpha}{x_{2k-1}}$

these gives:  $x_{2k+2} > x_{2k}$  and  $x_{2k+1} < x_{2k-1}$  for all  $k \in \mathbb{N}$ .

$$c) \lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$$

proof. Since  $\{x_{2n}\}$  and  $\{x_{2n-1}\}$  both are increasing and decreasing, respectively, bounded. Therefore both are convergent. Put  $L_1 = \lim_{n \rightarrow \infty} x_{2n}$  and  $L_2 = \lim_{n \rightarrow \infty} x_{2n-1}$ . Then,

$$L_1 = \lim_{n \rightarrow \infty} x_{2n+1} = \frac{\alpha + \lim_{n \rightarrow \infty} x_{2n}}{1 + \lim_{n \rightarrow \infty} x_{2n}} = \frac{\alpha + L_2}{1 + L_2} \text{ and similarly } L_2 = \frac{\alpha + L_1}{1 + L_1}$$

$$L_1 = \frac{\alpha + \frac{\alpha + L_1}{1 + L_1}}{1 + \frac{\alpha + L_1}{1 + L_1}} = \frac{\alpha + \alpha L_1 + \alpha + L_1}{2L_1 + \alpha + 1} = \frac{2\alpha + (\alpha + 1)L_1}{2L_1 + (\alpha + 1)} \Rightarrow L_1 = \sqrt{\alpha}$$

Similarly  $L_2 = \sqrt{\alpha}$ . Finally,

$$\sqrt{\alpha} \leq \liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n \leq \sqrt{\alpha}$$

implies  $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$

Section 3.20.

Suppose  $\{p_n\}$  is a Cauchy sequence in a Metric Space  $(X, d)$ .

If some subsequence  $\{p_{n_i}\}$  converges to  $p$ , then  $\{p_n\}$  converges to  $p$ .

Proof. Given  $\varepsilon > 0$ , take  $N_1 \in \mathbb{N}$  s.t

$$m, n \geq N_1 \Rightarrow d(p_m, p_n) < \frac{\varepsilon}{2}$$

and take  $N_2 \in \mathbb{N}$  s.t

$$k \geq N_2 \Rightarrow d(p_{n_k}, p) < \frac{\varepsilon}{2}$$

Then, for  $r \geq N = \max\{N_1, N_2\}$ ,

$$d(p_r, p) \leq d(p_r, p_{n_r}) + d(p_{n_r}, p) < \varepsilon.$$

( $N_r \geq N$  by definition)

Section 3.21.

Let  $(X, d)$  be a Complete Metric space, and  $\{E_n\}$  be a non-empty closed bounded subset of  $X$  s.t

$$E_1 \supseteq E_2 \supseteq \dots$$

If  $\text{diam } E_n \rightarrow 0$ , then  $\bigcap_{n=1}^{\infty} E_n$  is singleton.

Proof. If  $E = \bigcap_{n=1}^{\infty} E_n$  has two points,  $x, y \in E$ , then

$$d(x, y) \leq \text{diam } E_n = 0$$

which is contradiction. Thus, enough to show that  $E$  is not empty set.

For each  $k \in \mathbb{N}$ ,  $\bigcap_{i=1}^k E_i = E_k$  is not empty set, by assumption, therefore we can choose  $x_k \in E_k$ .

Then, the sequence  $\{x_k\}$  is a Cauchy sequence, because: let  $\varepsilon > 0$  be given.

Take  $N \in \mathbb{N}$  s.t  $\text{diam } E_N < \varepsilon$ . Then, for any  $m, n \geq N$ ,

$$d(x_m, x_n) \leq \text{diam } E_N < \varepsilon.$$

Therefore,  $\{x_k\}$  is Cauchy sequence in  $X$ . Since  $X$  is complete, put  $x = \lim_{n \rightarrow \infty} x_n$ .

Clearly, for any  $n \in \mathbb{N}$ ,  $\{x_k\}_{k=n}^{\infty} \subseteq E_n = \bar{E}_n$ , therefore  $x \in E_n$  for all  $n \in \mathbb{N}$ .

Finally,  $x \in E$ .